

16. DATA SUFFICIENCY

This section consists of problems in which a question on any topic such as Coding-Decoding, Puzzle Test, Blood Relations, Mathematical calculations etc., is put forward, followed by certain statements containing facts providing clues to solve the question. The candidate is required to find out which of the given statements is/are sufficient to answer the given question.

Ex. Out of six lectures of one hour each — **A, B, C, D, E** and **F** scheduled between 10 a.m. to 4 p.m., which one will be third ? **(Bank P.O. 1998)**

Statements : **I.** Lecture **F** is preceded by **A** and is followed by **C**.

II. There is only one lecture before **A** and there is no lecture after **B**.

Sol. From **I**, we get the sequence as **A, F, C**.

Also, it is given in **II** that there is only one lecture before **A**.

Clearly, **F** is the third lecture.

Thus, both **I** and **II** are required to answer the given question.

EXERCISE 16

Directions : Each question given below has a problem and two statements numbered **I** and **II** giving certain information. You have to decide if the information given in the statements is sufficient for answering the problem. Indicate your answer as

- (a) if the data in statement **I** alone are sufficient to answer the question;
- (b) if the data in statement **II** alone are sufficient answer the question;
- (c) if the data either in **I** or **II** alone are sufficient to answer the question;
- (d) if the data even in both the statements together are not sufficient to answer the question;
- (e) if the data in both the statements together are needed.

1. What is Reena's rank in the class ? **(Bank P.O. 1995)**

I. There are 26 students in the class.

II. There are 9 students who have scored less than Reena.

2. Who is the father of **M** ? **(S.B.I.P.O. 1994)**

I. **A** and **B** are brothers.

* **II.** **B**'s wife is sister of **M**'s wife.

3. What day is the fourteenth of a given month ?

I. The last day of the month is a Wednesday.

II. The third Saturday of the month was seventeenth.

4. Among four friends **A, B, C** and **D**, who is the heaviest ?

I. **B** is heavier than **A**, but lighter than **D**.

II. **C** is lighter than **B**.

5. It is 8.00 p.m., when can Hemant get next bus for Ramnagar from Dhanpur ?

- L Buses for Ramnagar leave after every 30 minutes, till 10 p.m.
- II. Fifteen minutes ago, one bus has left for Ramnagar. **(Bank P.O. 1897)**
6. In a certain code '13' means 'stop smoking' and '59' means 'injurious habit'. What is the meaning of '9' and '5' respectively in that code ?
- I. '157' means 'stop bad habit'.
- II. '839' means 'smoking is injurious'.
7. When is Manohar's birthday this year ? **(Bank P.O. 1993)**
- I. It is between January 13 and 15, January 13 being Wednesday.
- II. It is not on Friday.
8. On which day the flat was purchased by Rohan in 1996 ? **(Bank P.O. 1997)**
- I. Certainly before 18th December, 1996 but definitely not before 15th December, 1996.
- II. Certainly after 16th December, 1996 but not later than 19th December, 1996.
9. Is Arun taller than Sachin ?
- I. Dinesh is of the same height as Arun and Sachin.
- II. Sachin is not shorter than Dinesh.
10. Buses are always punctual in city X. How long, at the most, will Mr. Roy have to wait for the bus ?
- I. Mr. Roy has come to the bus stand at 9 A.M.
- II. There is a bus at 10 A.M. and possibly another bus even earlier.
11. The Chairman of a big company visits one department on Monday of every week except for the Monday of third week of every month. When *did* he visit the Purchase department ? **(Bank P.O. 1996)**
- I. He visited Accounts department in the second week of September after having visited Purchase department on the earlier occasion.
- II. He had visited Purchase department immediately after visiting Stores department but before visiting Accounts department. ~
12. How is D related to A ?
- I. B is the brother of A.
- II. B is D's son.
13. Gaurav ranks eighteenth in a class. What is his rank from the last ?
- I. There are 47 students in the class.
- II. Jatin who ranks 10th in the same class, ranks 38th from the last.
14. How many brothers does Tarun have ? **(Bank P.O. 1998)**
- I. Tarun's father has three children.
- II. Tarun has two sisters.
15. Rohit, Kajol, Tanmay and Suman are four friends. Who is the oldest among them ?
- I. The total age of Kajol and Tanmay together is more than that of Suman.
- II. The total age of Rohit and Kajol together is less than that of Suman.
16. How many new year's greeting cards were sold this year in your shop ?
- I. Last year 2935 cards were sold.
- II. The number of cards sold this year was 1.2 times that of last year.
- (S.B.L.P.O. 1997)**

17. In a certain code language, '297*' means 'tie clip button'. Which number means 'button' ?
(Bank P.O. 1995)
- I. In that language *926* means 'clip your tie*'.
II. In that language '175' means 'hole and button*'.
18. How is B related to A ?
I. A is B's sister.
II. D is the father of A and B.
19. Among four brothers — Anil, Pawan, Neeraj and Sahil, who is the heaviest ?
I. Anil and Pawan are of the same weight.
II. Pawan weighs more than Neeraj, but less than Sahil. (8.B.I.P.O. 1994)
20. Who is C's partner in a game of cards involving four players A, B, C and D ?
I. D is sitting opposite to A.
II. B is sitting right of A and left of D.
21. Is D brother of F ? (Bank P.O. 1994)
I. B has two sons of which F is one.
II. D's mother is married to B.
22. Which day of the last week did Satish meet Kapil at Kapil's residence ?
I. Kapil was out of town from Monday to Wednesday. He returned on Thursday morning.
II. On Friday night Satish telephoned his friend to inform that only yesterday he had got approval of Kapil after personally explaining to him all the details. (Bank P.O. 1998)
23. What is Gagan's age ?
I. Gagan, Vimal and Kunal are all of the same age.
II. Total age of Vimal, Kunal and Anil is 32 and Anil is as old as Vimal and Kunal together.
24. How is Rakesh related to Keshav ? (Bank P.O. 1996)
I. Tapan's wife Nisha is paternal aunt of Keshav.
II. Rakesh is the brother of a friend of Nisha.
25. Four plays A, B, C and D were staged one on each day on four consecutive days, but not necessarily in that order. Which day was the play C staged
-I. The first play was staged on 14th, Tuesday and was followed by play D.
II. Play A was not organised on 16th and there was a gap of one day¹ between A and B.
26- Manoj, Prabhakar, Akash and Kamal are four friends. Who among them is the heaviest ? (Bank P.O. 1994)
I. Prabhakar is heavier than Manoj and Kamal but lighter than Akash.
U. Manoj is lighter than Prabhakar and Akash but heavier than Kamal.
27. Hemant ranks tenth in a class. How many students are there in the class ?
I. His friend got 58th rank which is the last.
II. Hemant's rank from the last is 49th. (Bank P.O. 1998)
28. Vipin's and Javed's salaries are in the proportion of 4 : 3 respectively. What is Vipin's salary ?
I. Javed's salary is 75% that of Vipin's salary.
II. Javed's salary is Rs 4500.

29. At what time did Sonali leave her home for office ? (8JBJP.O.1997)
 I. Sonali received a phone call at 9.15 a.m. at her home.
 II. Sonali's car reached office at 10.15 a.m., 45 minutes after she left her residence.
30. How many sons does D have ? (S.B.I.P.O. 1994)
 I. A's father has three children.
 II. B is A's brother and son of D.
31. A, B, C, D and E are sitting in a row. B is between A and E. Who among them is in the middle ?
 I. A is left of B and right of D.
 II. C is at the right end.
32. How many gift boxes were sold on Monday ? (Bank P.O. 1997)
 I. It was 10% more than the boxes sold on the earlier day i.e. Sunday.
 II. Every third visitor to the shop purchased the box and 1500 visitors were there on Sunday.
33. What is the monthly salary of Praveen ?
 I. Praveen gets 15% more than Sumit while Sumit gets 10% less than Lokesh.
 II. Lokesh's monthly salary is Rs 2500.
34. How many pages of the book X did Robert read on Sunday ? (Bank P.O. 1998)
 I. The book has 300 pages out of which two-third were read by him before Sunday.
 II. Robert read the last 40 pages of the book on the morning of Monday.
35. In the last month the company decided to increase the cost of its mixer by 10 percent. What is the present price of the mixer ? (Bank P.O. 1996)
 I. The cost of mixer and juicer together was Rs 2850 a month ago.
 II. The amount of 10% increase on the mixer comes to Rs 220.
36. In a certain language, 'pit nac mit' means 'Yed pant shirt'. Which word means 'pant' in that language ?
 I. 'mit tim nac sir' means 'he wore red pant'.
 II. 'nee jic pit' means 'shirt is dirty'.
37. In a code, 'lee pee tin' means 'Always keep smiling'. What is the code for 'smiling' ?
 I. 'tin lut lee' means 'Always keep left'.
 II. 'dee pee' means 'Rose smiling'. (Bank P.O. 1993)
38. How many visitors saw the exhibition yesterday ? (Bank P.O. 1997)
 I. Each entry pass holder can take up to three persons with him/her
 II. In all, 243 passes were sold yesterday.
39. How much was the total sale of the company ?
 I. The company sold 8000 units of product A each costing Rs 25.
 II. This company has no other product line.
40. In what proportion would Raj, Karan and Altaf distribute profit among them ?
 I. R gets two-fifth of the profit.
 II. Karan and Altaf have made 75% of the total investment. (Bank P.O. 1997)
41. What will be the total weight of 10 poles each of the same weight ?
 I. One-fourth of the weight of a pole is 5 kilograms.
 II. The total weight of three poles is 20 kilograms more than the total weight of two poles.

42. Rsyeevs monthly salary is Rs 4000. What is Atul's monthly salary ?
I. Atul gets Rs 500 more than the average salary of his and R[^]jeev's.
II. Average of Ifeyeev's and Atul's salary is Rs 4500. **(Bank P.O. 1993)**
43. Among five friends who is the tallest ?
I. D is taller than A and C.
II. B is shorter than E but taller than D.
44. What is the price range of ordinary wall clocks ? **(Bank P.O. 1998)**
I. The price range of ordinary wrist watches of company X is Rs 400 to Rs 600.
II. The price range of ordinary wall clocks of company X is 50 percent that of their ordinary watches.
45. What is the amount of rice exported from India ?
I. India's export to America is 80,000 tonnes and this is 10% of the total rice exports.
II. India's total export tonnage of rice is 12.5% of the total of 1.9 million tonnes.
46. How much amount Ronnie required to pay for the new car in the buy-back scheme ? **(83.LP.O.1997)**
I. The cost of the new car was three times the cost price of his old car.
II. His old car was valued at Rs 25000 under buy-back scheme.
47. How many votes did candidate X receive in the City Cooperative bank's director's election ? **(Bank P.O. 1997)**
I. Candidate X got 17 percent of the votes that were cast.
II. Four-fifth of the 1000 eligible voters cast their votes.
48. What is Manohar's birthday ?
I. Manohar's father was bom on 27th May, 1948.
II. Manohar is 25 years younger than his mother.
49. On which day in April is Gautam's birthday ? **(Bank P.O. 1994)**
I. Gautam was born exactly 28 years after his mother was born.
II. His mother will be 55 years 4 months and 5 days on August 18 this year.
50. Total money with Naresh and Ajay is 28 percent of that with Usman. How much money is Ajay having ? **(Bank P.O. 1996)**
I. Usman has got Rs 75000.
II. The ratio of money of Naresh to money held by Ajay is 1 : 3.
51. What time did the train leave today ?
I. The train normally leaves on time.
II. The scheduled departure is at 14 : 30.
52. On which day in January, Subhas left for Germany ?
I. Subhas has so far spent 10 years in Germany.
II. Subhas friend ^Vnil left for Germany on 15th February and joined Subhas 20 days after Subhas' arrival.
53. In which year was Rahul born ?
I. Rahul at present is 25 years younger to his mother.
II. Rahul's brother, who was born in 1964, is 35 years younger to his mother.
54. What is the area of this plot ? **(S.B.I.P.O.1997)**
I. The perimeter of the plot is 208 metres.
II. The length is more than the breadth \y 4 metres.

55. How many speeches were delivered in the two days' programme ?
- I. 18 speakers were invited to give at least one speech, out of which one-sixth of the speakers could not come.
 - II. One-third of the speakers gave two speeches each. (Bank P.O. 1997)
56. What is the birthdate of Sonali's mother ? (Bank P.O. 1995)
- I. Sonali's father remembers that his wife's birthday is after 20th and before 23rd February.
 - II. Sonali's brother remembers that their mothers birthday was after 21st but before 25th February.

Directions (Questions 57 to 64) : *In each of the following problems, there is one question and three statements I, II and III given below the question. You have to decide whether the data given in the statements is sufficient to answer the question. Read all the statements carefully and find out that probable pair which can be sufficient to answer the question. Any one such alternative which contains the statement or a pair of statements sufficient to answer the question, will be your answer. For example, if only statement I is sufficient to answer the question, then statements I and II together should not be accepted as answer to the question. Remember out of the three statements, each of them alone can also be sufficient to answer the question. In such cases for example, your answer should be taken as Only I or Only II or Only III and not Only I.*

57. Pankaj is younger than Sunita and Rupali is older than Tom. Who among them is the oldest ?
- I. Rupali is older than Pankaj.
 - II. Sunita is older than Rupali.
 - III. Tom is the youngest among all.
- (a) Only II (6) Only III (c) I and II together
(d) I, II and III all together (e) None of these
58. What does 'come' represent in a code language ? (S.B.I.P.O.1997)
- I. 'pit na tac' means 'come and go' in that code language.
 - II. 'ja ta da' means 'you are good' in that code language.
 - III. 'na da rac' means 'you can come' in that code language,
- (a) I and II together (6) II and III together (c) I and III together
(d) I, II and III all together (e) None of these
59. Five persons — A, B, C, D and E are sitting in a row. Who is sitting in the middle ?
- I. B is between E and C.
 - II. B is to the right of E.
 - III. D is between A and E.
- (a) I and II together (6) II and III together (c) I and III together
(d) I, II and III together (e) None of these
60. What is the total monthly salary of Vasu ? (S.B.I.P.O.1995)
- I. Vasu's basic salary is Rs 100 more than Rajan's salary who also serves in Vasu's company.

- II. Other allowances drawn by R^jan besides his basic salary are Rs 2000 per month which is Rs 50 less than Vasu's salary.
- III. Ryan's basic salary is Rs 1550 per month.
- (a) II only (6) II and III together (c) I and II together
(d) I and III together (e) I, II and III together
61. In which year was Sanjay born ?
- I. Sanjay is six years older than Gopal.
II. Gopal's brother was born in 1982.
III. Sanjay's brother is two years younger than Gopal's brother who was eight years younger than Gopal.
- (a) I and II together (6). II and III together (c) I and III together
(d) I, II and III together (e) None of these
62. Who among Siddhartha, Nikunj, Vipul and Mukul is the youngest ?
- I. Vipul is younger than Mukul but older than Siddhartha and Nikunj.
II. Mukul is the oldest.
III. Siddhartha is older than Nikunj. **(8.B.I.P.O.1997)**
- (a) Only I (ft) I and II together (c) II and III together
(d) I and III together (e) None of these
63. Four subjects — Physics, Chemistry, Mathematics and Biology were taught in four consecutive periods of one hour each starting from 8.00 a.m. At what time was the Chemistry period scheduled ? **(S.B.I.P.O. 1995)**
- I. Mathematics period ended at 10.00 a.m. which was preceded by Biology.
II. Physics was scheduled in the last period.
III. Mathematics period was immediately followed by Chemistry.
- (a) Only I (6) Only I or Only II (c) Only II
(d) II and III together (e) I and II together or I and III together
64. How many sons does X have ?
- I. Q and U are brothers of T.
II. R is sister of P and U. ,
III. R and T are daughters of X.
- (a) I and II only (6) II and III together (c) I, II and III together
(d) I, II and III together are not sufficient (e) None of these

ANSWERS

1. *it*): From I and II, we conclude that there are 16 students above Reena in rank. Thus, Reena's rank is 17th in the class. So, both the statements are necessary.
2. *(tf)*: From II, we conclude that B is the brother-in-law of M. So, even from both the statements*, we can't find out who is the father of M.
3. (6): Statement II reveals that 17th was a Saturday and therefore, 14th was Wednesday. So, only statement II is needed
4. (e): From I, we have $A < B < D$*(i)*
From II, we have $C < B$. „ ...*(#)*
Combining (i) and (ii), we can conclude that D is the heaviest. So, both the statements are needed
5. (e): II reveals that the previous bus had left at 7.45 p.m. As given in I, the next bus would leave after 30 minutes i.e. at 8.15 p.m.

6. (c) : means 'injurious habit' and '157' means 'stop bad habit' (from I). Thus, the common code number '5' stands for common word 'habit'. So, '5' represents 'injurious'. Hence, I is sufficient. Also, '59' means 'injurious habit' and '839' means 'smoking is injurious'. Thus, the common code number '9' stands for common word 'injurious'. So, '59' represents 'habit'. Thus, II is also sufficient.
7. (a): From statement I, we conclude that Manohar's birthday is on January 14, which is Thursday, this year. So, only I is needed.
8. (e): From statement I, we conclude that Rohan purchased the flat between 15th and 18th December *i.e.* on 16th or 17th December.
From statement II, we conclude that the flat was purchased between 16th and 19th December *i.e.* on 17th or 18th December.
The day common to above two groups is 17th December. So, Rohan purchased the flat on 17th December.
Thus, both I and II are needed.
9. (a): From statement I, we can conclude that Dinesh, Arun and Sachin are of the same height. So, Arun is not taller than Sachin. Thus, only statement I is sufficient to answer the question.
10. (e) ; From both the given statements, we find that Mr. Roy reached the bus stand at 9 A.M. and a bus is sure to arrive at 10 A.M. So, Mr. Roy has to wait for at the most one hour.
11. (a) : From statement I, we can conclude that the Chairman visited Purchase department on Monday of the first week of September. So, I alone is sufficient. The time of visit of no department is mentioned in II, which is, therefore, insufficient.
12. (e): From both the statements together, we find that D is the father of B and B is the brother of A. So, D is the father of A. Thus, both the given statements are needed.
13. (c): To find the rank from other end, we need to know the total number of students in the class. So, I is sufficient.
Also, from II, we can conclude that there are $(10 \times 38 - 1) = 47$ students in the class. So, II alone is also sufficient.
14. (e): From both the statements together, we find that Tarun's father has three children — Tarun and his two sisters. This means that Tarun has no brother. So, both I and II are needed.
15. (d): From given statements, we find that :
 $K > T$ and $R < K < S$.
Thus, who is oldest is not known.
16. (e): From both the given statements, we find that the number of cards sold this year is $(2935 \times 1.2) = 3522$.
So, both I and II together are required.
17. (c): Comparing the information in the question with statement I, we find that '2' and 'A' are the codes for 'tie' and 'clip'. So, '7' represents 'button'. Thus, I alone is sufficient. Again, comparing the information in the question with II, we find that the common code number 'T' stands for the common word 'button'. Thus, II alone also is sufficient.
18. (d): From statements I and II together, we can conclude only that either B is the sister or brother of A. So, even from both the statements, the exact relation cannot be known.
19. (r): From I, we have : $A = P$.
From II, we have : $N < P < S$.
Combining the above two, we have : $N < A = P < S$.
Sahil is the heaviest.
So, both I and II are required.
20. (c): Clearly, each of the given statements shows that B is sitting opposite to C or B is the partner of C.

21. (e) : From I, we conclude that F is the son of B.
From II, we conclude that B's wife is D's mother.
This means that D and F are the sons of B and D is the brother of F.
So, both I and II are required.
22. (6): Statement II reveals that Satish met his friend Kapil on the day prior to Friday i.e. Thursday. Thus, only II is needed.
23. (e): As given in statements I and II, we have
 $G = V = K, V + K + A = 32$ and $A = V + K$.
If Putting $V + K = A$ in second, we get $2A = 32$ or $A = 16$.
Thus, $V + K = 16$ and $V = K$. So, $V = K = 8$. Thus, R = 8.
Hence, both the statements are needed.
24. (d): Clearly, both the statements together are not sufficient to answer the question.
25. (e): From both the statements, we obtain the following sequence :

14th	15th	16th	17th
Tuesday	Wednesday	Thursday	Friday
in A	D	B	C

- Thus, both I and II are required.
26. (a): From I we have. $P > M, P > K, A > P$.
From II we have, $M > K, P > M, A > P$.
Combining the above two, we have : $A > P > M > K$
Thus, Akash is the heaviest.
So, both the statements are required
27. (c): I reveals that 58th is the last rank in the class. This means that there are 58 students in the class. So, I alone is sufficient.
Also, from II, we find that Hemant's rank in the class is 10th from the top and 49th from the last. So, there are $(10 + 49 - 1) = 58$ students in the class.
Thus, II alone is also sufficient.
28. (6): Statement I is merely an interpretation of the information contained in the question. However, Vipin's salary can be determined from statement II as follows :
Let Vipin's and Javed's salaries be x and $3x$ respectively.
Then, $3x = 4500$ or $x = 1500$.
Vipin's salary $= 4x = \text{Rs } 6000$.
Thus, II alone is sufficient.
29. (6): From statement II, we conclude that Sonali left her home 45 minutes before 10.15 a.m. i.e. 9.30 a.m.
Thus, only II is needed.
30. (d): From both I and II together, we can conclude that A and B are the children of D, but the sex of A and the third child of D is not known. So, both the statements together are also not sufficient to answer the question.
31. (e): Clearly, we have the order : A, B, E.
From I, we have the order : D, A, B, E.
From II, we get the complete sequence as D, A, B, E, C.
Clearly, B is in the middle.
So, both I and II are required.

(1500

g i.e. 500 boxes were sold on Sunday.

From I, we find that number of boxes sold on Monday $= 500 + 10\% \text{ of } 500 = 550$.

33. *ie*): From both the given statements, we find that :
 Pmvecn s salary - 115% of (90% of Rs 2500) » Rs 2587 50
 So. both I and II are required.
34. (e): From I and II. we find that Robert read $300 * \frac{1}{3}$ i.e. 200 pages before Sunday and
 the last 40 pages on Monday. This means that he read $(300 - (200 - 40))$ i.e. 60 pages
 on Sunday.
35. (6) : The present price of the mixer can be calculated from statement II as follows :
 Let the earlier price be Rs x .
 Then, 10% of x - 220 or $x - 2200$.
 Present price = $(2200 + 220)$ - Rs 2420.
 So, only II is needed.
36. *id*) : Clearly, from each of the statements, we find that the code for 'pant' is either 'mit'
 or 'nac'. So, none of them is sufficient to answer the question.
37. (c) : Comparing the information in the question with I, we find that 'tin' and W are the
 codes for 'always' and 'keep'. So, 'pee' represents 'smiling'. Thus, I alone is sufficient.
 Again, comparing the information in the question with II, we find that the common code
 word 'pee' stands for the common word 'smiling'. Thus, II alone is also sufficient.
38. (d) : From the two statements, we find that maximum (243×3) i.e. 729 visitors saw the
 exhibition. But the exact number cannot be determined.
39. (e) : From statements I and II together, we find that company sells only product A and
 total sale of product A = Rs (8000×25) - Rs 200000, which is also the total sale of
 the company.
 So, both I and II are required.
40. (d): Even both the statements together are not sufficient to answer the question.
41. (c): From I. we find that the weight of one pole is (5×4) i.e. 20 kg and so the weight of
 10 poles is 200 kg. Thus, I alone is sufficient.
 From II. we have : (weight of 3 poles — weight of 2 poles) = 20 kg
 or weight of one pole = 20 kg.
 So. weight of 10 poles = 200 kg.
 (Thus. II alone is also sufficient.
42. (c) : From I, we have :

$$A - 4000 = 1000 \text{ or } A - 5000$$

 i.e. Atul's monthly salary = Rs 5000.
 From II, we have :

$$A - 4000 = 9000 \text{ or } A = 13000$$

 Thus, either I or II is sufficient.
43. (e): From I, we have : $D > A$. $D > C$.
 From II, we have : $E > B > D$.
 Combining the above two, we get : $E > B > D > (A \text{ and } C)$.
 So, E is the tallest.
 Clearly, both the statements are needed to answer the question.
44. (e) : From both the statements together, we can conclude that the price range of ordinary
 wall clocks is Rs 200 to Rs 300. Thus, both I and II are required.
45. (6) : From statement II. we conclude that the total amount of rice exported from India
 is (12.5% of 1.9 million tonnes)
 So. the answer to the given question can be derived from statement II alone.

46. (r): From both the statements together, we conclude that the price of the new car is Rs (25000×3) i.e. Rs 75000
So, Ronnie had to pay Rs $(75000 - 25000)$ i.e. Rs 60000.
Thus, both I and II are required

47. (*): From statement II, we conclude that number of votes cast - $\frac{1}{2} \times 1000 = 500$.
From statement I, we conclude that votes polled by candidate X = 17% of 800 = 136.
48. id) : Clearly, even both the statements together do not reveal Manohar's birthday.
49. ie): Clearly, the birthday of Gautam's mother can be found out from II and then Gautama birthday can be determined using the fact given in I. Thus, both the statements are required.
50. ie): From I, we find that

Total money with Naresh and Ajay = Rs $\frac{1}{2} \times 75000 =$ Rs 21000.

From II, we find that money with Ajay = Rs $\frac{1}{3} \times 21000 =$ Rs 15760.

So, both I and II are required.

51. id) : Clearly, even both the statements together do not reveal the exact time of departure of the trail) today.
52. id) : Clearly, even from both the given statements, we cannot conclude the exact date of Subhas* leaving for Germany.
53. ie): From both the given statements, we find that Rahul is $(35 - 25) = 10$ years older than his brother, who was born in 1964. "So, Rahul was born in 1954. Thus, both the given statements are needed to answer the query.
54. (e) : The area of the plot can be determined from both I and II as follows :
Let the breadth of the plot be x metres.
Then, length = $(x + 4)$ metres.
Thus, perimeter = 208 metres $\Rightarrow 2(x + 4 + x) = 208$
 $\Rightarrow 4x = 200 \Rightarrow x = 50$.
So, length = 54 metres, breadth = 50 metres.
Area of the plot = $(54 \times 50) \text{ m}^2 = 2700 \text{ m}^2$.
Hence, both I and II are required.

55. (e) : From I, we find that number of speakers who attended the programme = $18 - \frac{1}{3}$ of 18
 $= 15$.

(I)
From II, we find that $\frac{1}{5} \times 15 = 3$ i.e. 5 speakers gave 2 speeches each, while each of the remaining 10 speakers delivered only one speech.
So, total number of speeches delivered = $(5 \times 2 + 10 \times 1) = 20$.

56. ie) : From I, we find that the birthday of Sonali's mother falls on 21st or 22nd February.
From II, we find that the birthday of Sonali's mother falls on 22nd, 23rd or 24th February.
The day common to the above two groups is 22nd February. So, the birthday falls on 22nd February.
Thus, both I and II are required.

57. ia): Given : $P < S, T < R$... (j)
From I, we have : $P < R$... (ii)
From II, we have : $R < S$... (fli)

